

News Sentiment - A New Yield Curve Factor

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Abstract

We show that sentiment from newspaper articles is able to explain and predict movements in the term structure of US government bonds. This effect is stronger at the short end of the curve, coinciding with greater volatility and investors' need to continually reassess the Fed's reaction function. Facing such uncertainty, market participants rely on news and sentiment as a central element in their decision-making process. In light of this dependence, we propose a new yield curve factor - news sentiment - which is distinct from the three established yield curve factors (level, slope and curvature) as well as from fundamental macroeconomic variables.

JEL-Codes: G02, G11, G12, G17

Keywords: Behavioral Finance, News Sentiment, Duration, Yield Curve

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The impact of news sentiment on equity markets has been well established in the literature (Tetlock, 2007; Garcia, 2013). In this paper, we focus on two main research questions to broaden the literature on news sentiment: does the sentiment derived from newspaper articles impact sovereign bond investors and thus the U.S. government bond yield curve? Does the significance of this effect vary depending on the duration of the bond taken into consideration? We explore these questions using yield curve data for U.S Treasuries as well as a novel news sentiment variable. In this context, the interplay between sovereign bond markets and news sentiment has yet to be examined. We therefore aim to close this gap in the literature.

We find that news sentiment offers explanatory power for shorter maturities of the US Treasury yield curve, in line with investors' need to continually reassess the reaction function of the Fed. Therefore, we identify a novel yield curve factor - news sentiment - which is distinct from the traditional level, slope and curvature measures. The remainder of the paper is organized as follows: section 1 surveys the existing literature, section 2 explores the data used, section 3 discusses the empirical results, and section 4 concludes.

1 News Sentiment and Yields

Surprises in news articles can move financial markets. In particular, the influence of sentiment in news releases on equity markets has been examined in greater detail. For instance, Tetlock (2007) investigates the relationship

between a daily news column in the *Wall Street Journal* and subsequent stock market returns. The author shows that media pessimism predicts future negative stock market returns and can thus move aggregate quantities. Building on this work, Garcia (2013) demonstrates that the predictive power of sentiment derived from media content (in this case extracted from The New York Times) is particularly potent in periods of recession. Similar results also apply on a firm specific level in the cross section, as Dzielinski (2011) shows.

As the relevance of news sentiment in equity markets has moved into focus, several global providers have emerged which generate automated databases. Thomson Reuters offers one such service called Thomson Reuters News Analytics (TRNA), in which news items are rated in terms of their underlying sentiment. An advanced natural language processing algorithm is used to code news items in line with various topic classifiers, offering a sophisticated approach of rating sentiment in newspaper articles. Leinweber and Sisk (2011) and Uhl et al. (2015) use this dataset in order to show a longer-run effect of sentiment in news on equities. We build on these findings in our study.

Whereas several studies show that news and their derived sentiment can influence and forecast stock market activity,¹ other studies have related news sentiment to different asset classes, but not to US yield curve movements.² Thus, it has not been shown that news sentiment explains US Treasury yield curve movements.

¹See Antweiler and Frank (2004), Cao and Wei (2005), Edmans et al. (2007), Hirshleifer (2001), Hirshleifer and Shumway (2003), Kamstra et al. (2003), Maier (2005), Mullainathan and Shleifer (2005), and Gross-Kluschmann and Hautsch (2011), among others.

²See Smales (2016) who examines news sentiment and credit spreads, Uhl (2017) considers news sentiment and foreign exchange, and Smales (2014) looks at news sentiment and commodity markets.

Prior research on sovereign bond markets has focused on capturing the effect of macroeconomic data releases. Edison (1996) establishes the importance of inflation and employment related releases (CPI and non-farm payrolls) for Treasury yields. Measures of output and inflation have a significant influence on sovereign interest rates.³ The term structure is furthermore found to be sensitive to changes in inflation uncertainty, as seen in the context of current central bank monetary policy frameworks (Wright (2011)). Such uncertainty drives investors to update their expectations more swiftly as a response to news (David and Veronesi (2004)). A link exists between uncertainty regarding the future development of fundamentals and the volatility of Treasury bond yields. Thus, the relationship between macroeconomic news and yields is not constant over a given period. Goldberg and Grisse (2013) provide evidence for time variation in the response of the 10-year US Treasury yield to economic data, as a reweighting of output and inflation occurs in the Federal Reserve reaction function. Markets appear to take into consideration that the central bank is subject to a latent financial stability objective.⁴

While there has been a considerable effort devoted to purely fundamental approaches, there are more studies which have begun to take into consideration measures of investor sentiment, employing more differentiated behavioral analyses. Gade et al. (2013) shows that sentiment associated with political statements is highly significant in explaining sovereign peripheral European bond market spreads during the Eurozone debt crisis. The dynamic nature of

³See Edison (1996), Goldberg and Leonard (2003), Gürkaynak et al. (2005), and Goldberg and Leonard (2003).

⁴Several other papers have examined the impact of monetary policy on yields, including Chun (2011).

this relationship between sovereign bond yields and economic news releases is established in Goldberg and Grisse (2013). Therefore, a degree of time variance exists, in line with a reweighting of the importance of output and inflation factors on behalf of the Federal Reserve.

This study considers whether news sentiment can explain and predict movements in US government bond yields along the yield curve. In this context, news sentiment may reflect new information regarding fundamentals which has yet to be incorporated into prices, pure sentiment referred to as noise, or a mixture of the two (Tetlock (2007)). Yield movements induced by noise should have no lasting effect. Given a novel information channel, interest rates should shift in a more permanent fashion. Both empirical and theoretical work has shown that in such cases over- and underreaction are prone to occur as investors misinterpret new information.⁵ Such momentum stems from two central psychological biases, namely overconfidence and self-attribution, in line with the work of Daniel et al. (1998). As such, investors tend to over- (under-) weight the accuracy of private signals which spurs over- (under-) reaction. In this study, it may relate to a perceived superior ability to interpret news and its implications for sovereign bond markets. Once said signal is public, it is followed by an over- (under-) reaction to such information. These behavioral biases are enhanced by the presence of uncertainty and volatility, and are thus more likely to be present at the short end of the curve.

Given the pronounced effect of news sentiment on stock returns, the application of such data in the realm of government bond yields appears to be

⁵See Barberis et al. (1998), Gilbert et al. (2010), and Hong and Stein (1999), among others.

a logical extension. An interesting point in this undertaking is to extend the findings from the literature that considered news sentiment and equities to news sentiment and yields.

2 Data

This section examines the data, its characteristics as well as the construction methodology for all variables considered in the modeling framework. This involves news sentiment, several macroeconomic control variables, US Treasury yields along the curve as well as yield curve factors.

2.1 News Sentiment

To shed more light on the possible behavioral drivers of yield curve movements, we examine the sentiment extracted from articles spanning the entire Thomson Reuters News Analytics universe. Since Thomson Reuters is a leading global news agency, we ensure that the news coverage is as exhaustive as possible and is frequently used by financial professionals for information regarding financial markets.

Our database is referred to as Thomson Reuters News Analytics (TRNA) and captures both positive and negative sentiment pertaining to media content. Initially, a fixed set of keywords with positive and negative connotation are determined using the so-called “bag-of-words” approach. A learning algorithm is then applied, which was constructed from the analysis of input from several hundred financial market experts and economists. A fixed set of articles were provided to these individuals, who assigned positive or negative

sentiment scores based on their opinion of the content. The resulting sentiment scores were then used as input into the adaptive algorithm, leading to a coding classifier that is able to identify positive (1), neutral (0) and negative (-1) sentiment of each article.⁶ This approach is consistent with the work of Tetlock (2007), as the data is generated from the whole text body of each newspaper article. Furthermore, the data is clean in terms of data quality, availability and construction.

This dataset is available in the form of high frequency tick data related to a wide variety of topic classifiers.⁷ For the purpose of this study, news sentiment items are filtered for topics relating to political, debt, and monetary policy news. As shown in Gade et al. (2013), sentiment regarding political communication can have a significant impact on yields. Additionally, a subjective interpretation of the course of future monetary policy and central bank objectives induces bond market movements, as noted in Goldberg and Grisse (2013). As the level of debt is directly tied to the creditworthiness of a sovereign, the news sentiment pertaining to this factor is also included.

Figure 1 lists the relevant topics (debt, monetary, and political) taken into account for this study, from which we construct the news sentiment variable. Table 1 shows the summary statistics of the three news sentiment components and the total news sentiment variable. As in Uhl et al. (2015), weekends

⁶See Thomson Reuters News Analytics, <http://thomsonreuters.com/content/dam/openweb/documents/pdf/tr-com-financial/news-analytics-product-brochure-oct-2010.pdf>, last accessed 7 September 2016.

⁷The topics range from disaster and terrorist news, war, sports, stock-specific, political, debt news to monetary policy news, categorized into topic codes. See *Reuters Codes – A quick guide*, available at https://customers.reuters.com/training/trainingCRMdata/promo_content/ReutersCodes.pdf, last accessed 7 September 2016.

are excluded to ensure a lower likelihood of overestimation of sparse data, as financial news are generally less frequent over the weekend and to a large degree repetitive. The 5-day weekly data are then aggregated to monthly data in order to examine the effect over a longer-time span (as opposed to high-frequency) and in order to being able to include monthly macroeconomic data as control variables in the regressions. Our data considers the monthly average of the news sentiment scores, as opposed to end-of-month data, in order to include all relevant fluctuations (given our hypothesis that news sentiment can persist over a longer period of time). Over 400,000 Reuters news items per month were coded for sentiment from January 2003 to February 2014, totaling 134 monthly observations.⁸

2.2 US Treasury Yields and Macroeconomic Variables

For our analysis, the 1- and 3-month, 1-, 2-, 3-, 5-, 7-, 10- and 30-year generic US Treasury yields as well as the 10-year minus 2-year yields are taken into consideration. All yield curve data were retrieved from Bloomberg on a monthly basis (end-of month data). In order to control for various other factors which may explain and predict yield movements and in line with the literature described above, we have included several macroeconomic control variables. In line with the work of Edison (1996), we focus on changes in inflation and economic activity. This includes the year-over-year measure of consumer price inflation (CPI). Given a positive change in inflation dynamics, we expect rates to move in the same direction as growth expectations move higher, as the

⁸The database software used for all operations is kdb+. The end date of February 2014 is given because of data availability.

possibility of a rate hike by the Federal Reserve increases, as Edison (1996) points out. In terms of output measures, monthly non-farm payrolls provide an overview of the state of the labor market, which is part of the mandate of the Federal Reserve. The Institute for Supply Management (ISM) Manufacturing Index takes into account employment, new orders, production, inventories and supplier deliveries in 18 different industries.⁹ A value over 50 is generally considered to indicate an expanding economy, while prints under this mark point to a contraction. Both measures of output should have a positive relationship with interest rates in the same manner as inflation. The aforementioned data were downloaded from Bloomberg on a monthly basis from January 2003 to February 2014. ISM Manufacturing is considered a leading indicator and is normally released in the first week of each month. However, CPI and non-farm payrolls consider the previous month, and are thus published with a lag. Due to this publication delay, we have lagged these two particular variables to obtain a model that is realistic and timely. We have also lagged the 10-year minus 2-year yield curve factor.

⁹See The Institute of Supply Management, Manufacturing ISM Report on Business, <http://www.ism.ws/ismreport/mfgrob.cfm>, last accessed 7 November 2014.

3 Empirical Findings

3.1 News sentiment, macroeconomic variables and the US Treasury term structure

We explore the ability of news sentiment to explain movements in the term structure of US sovereign bonds. We consider US Treasuries with maturities ranging from 1 month to 30 years in our analysis, performing a series of simple OLS regressions as follows:

$$\Delta r_t = k + ns_t + \varepsilon_t, \quad (1)$$

where k is a constant, r_t refers to the 1- month to 30-year US Treasury yield, respectively, ns_t refers to news sentiment, and ε_t to the error term. All Treasury yields are taken in differences as these are non-stationary. News sentiment is estimated in levels, as the Augmented Dicky Fuller test allows the null hypothesis of a unit root to be rejected at the 1%-level.

The regression results are displayed in table 2. In particular, we find that the news sentiment factor is statistically significant with a positive coefficient across all maturities. An increase in the sentiment pertaining to media content is thus associated with a rise in interest rates. However, this does not amount to a parallel shift in the term structure, as the level of significance declines with increasing maturity. While the shorter-term yields are significant at the 1% level, the 7-, 10-, and 30-year yields are only statistically significant at the 10%-level. In addition, the Durbin Watson statistic is close to two, which

is in line with minimal autocorrelation while the standard error of regression remains small.

In order to control for the above mentioned macroeconomic and yield curve factors, we extend the OLS regression as follows:

$$\Delta r_t = k + ns_t + W_t + \varepsilon_t, \quad (2)$$

where W_t includes the control variables, which are the ISM Manufacturing Index, Consumer Price Index (yoy), Non-farm payrolls and the 10-year minus 2-year yield curve factor. Table 3 shows the results. It can be seen that the news sentiment variable remains statistically significant from 1-month to 7-year yields. For 10-year and 30-year yields it is not statistically significant anymore. The macroeconomic control variables show an opposing picture, as some tend to become more statistically significant the higher the maturity of the US Treasury. In particular, the ISM Manufacturing Index and the Consumer Price Index are statistically significant. Non-farm payrolls are not statistically significant. The US Treasury 10-year minus 2-year yield curve factor is more statistically significant at the front end of the curve. These findings are in line with our hypothesis, which is that news sentiment influences investors more at the short end of the yield curve and macroeconomic factors become more relevant at the long end of the yield curve.

As a further empirical validation and to examine predictability of news sentiment, we generate one-step ahead out of sample forecasts for all maturities based on equation (2).¹⁰ Additionally, we run forecasts in the same form as in

¹⁰Please refer to Appendix A.1 for technical details regarding the one-step-ahead out-of-

equation (2) but just without news sentiment as independent variable in order to being able to compare the results and thus check for the predictive power of news sentiment. As can be seen from the range of statistics provided in tables 4 and 5, the forecast errors are larger in the model without news sentiment (table 5), particularly for shorter maturities. The R-squared and adjusted R-squared values confirm as well that news sentiment has higher predictive power at the short end of the curve, as the R-squared values converge for the two regressions with and without news sentiment for 10-year and 30-year yields. These findings are in line with the regression results, and they confirm that news sentiment adds not only explanatory power but also predictive power for US Treasury yields, but mainly at the short end of the curve.

3.2 News sentiment - a new yield curve factor

Given the breadth of literature that has shown the importance of the three traditional yield curve factors "level", "slope" and "curvature" in explaining the term structure, we decompose the yield curve across the various considered maturities with a principal component analysis (PCA), as can be seen in Figure 2.

Existing yield curve literature has shown that the "level", "slope" and "curvature" factors are able to explain the majority of the variation of the Treasury yield curve. Numerous studies have since come to associate various variables to these factors, although there still remains much debate over the exact mechanisms in play. For instance, Evans and Marshall (2007) associate shocks to economic activity and price levels with "level" effects across the sample forecasts.

yield curve. The effect is found to be the strongest for stocks which induce inflation and output to move in the same direction. Dewachter and Lyrio (2006) also find that the standard “level” factor is highly correlated to long-run inflation expectations, the “slope” factor captures temporary business cycle conditions, while the “curvature” factor represents an independent monetary policy factor. Interestingly, Ang and Piazzesi (2003) note that these “labels stand for the effect the factors have on the yield curve rather than describing the economic sources of the shocks.” In their study, the authors show that “slope” and “curvature” factors can be related to macro factors, while the “level” factor survives largely intact when macro variables are incorporated.

The news sentiment factor is found to influence the yield curve in a very different manner from the other yield curve factors, as can be seen in figure 2. It is evident that the news sentiment factor is almost perfectly negatively correlated to the “slope” factor. Based on this, we conclude that “sentiment driven” investors are predominantly present at the short end of the curve while more “fundamental” investors select longer dated maturities. This coincides with the findings of Diebold and Li (2006) who show that volatility due to uncertainty is greater at the short end of the curve. Tasked with the difficult undertaking of interpreting the reaction function of the Federal Reserve, our results indicate that investors appear to rely on the sentiment contained in newspaper articles about topics ranging from monetary policy, political circumstances, and macroeconomic releases.

3.3 Robustness checks

In order to test for robustness of our empirical findings, we regress news sentiment with 1- to 5-year Fama-Bliss discount bond yields according to Fama and Bliss (1987), as shown in table 6. All news sentiment coefficients are statistically significant and have a positive coefficient sign, confirming our previous findings that news sentiment exhibits explanatory and predictive power for US Treasury yields.

To ensure that the news sentiment measure is not merely a reflection of these macroeconomic variables, we have regressed the lagged macroeconomic control variables on news sentiment and found no statistical significance in any of these regressions. Furthermore, we run regressions of the yield curve factors, for which we constructed time series for our sample based on Diebold and Li (2006), as outlined in Appendix A.2 and shown in table 7. None of the yield curve factors can explain news sentiment in a statistically significant manner. Therefore, we can assume that news sentiment contains separate information to the macroeconomic and yield curve factor control variables.

Thus, facing heightened uncertainty for shorter dated maturities (much of which is concerned with correctly interpreting the Fed reaction function), investors are increasingly attune to the sentiment contained in newspaper articles. As news sentiment factors into the decision making process, it creates a feedback mechanism which further enhances volatility. Hillert et al. (2014) make a similar finding for equities, showing that enhanced coverage by news outlets can exacerbate investor biases. This effect depends on the tone of reporting and is stronger in states of greater uncertainty. We therefore note that

we have found a new yield curve factor: news sentiment.

4 Conclusion

We establish that the sentiment contained in newspaper articles is able to explain and predict movements in the term structure of US sovereign bonds. Given the results of our empirical analysis, it appears that news sentiment contains new information, as opposed to pure noise. This effect is strongest at the short end of the curve and is distinct from the traditional "level", "slope" and "curvature" factors. Facing heightened volatility and uncertainty for shorter dated maturities (due to the need to correctly estimate the Federal Reserve reaction function), investors are increasingly "sentiment driven". Meanwhile, the long end of the curve is populated by more "fundamental" investors, which base their decisions on macroeconomic data. We therefore identify a novel yield curve factor called "news sentiment."

References

- Ang, A. and Piazzesi, M. (2003). A no-arbitrage vector autoregression of term structure dynamics with macroeconomic and latent variables. *Journal of Monetary economics*, 50(4):745–787.
- Antweiler, W. and Frank, M. Z. (2004). Is all that talk just noise? the information content of internet stock message boards. *The Journal of Finance*, 59(3):1259–1294.
- Barberis, N., Shleifer, A., and Vishny, R. (1998). A model of investor sentiment. *Journal of financial economics*, 49(3):307–343.
- Cao, M. and Wei, J. (2005). Stock market returns: A note on temperature anomaly. *Journal of Banking & Finance*, 29(6):1559–1573.
- Chun, A. L. (2011). Expectations, bond yields, and monetary policy. *Review of Financial Studies*, 24(1):208–247.
- Daniel, K., Hirshleifer, D., and Subrahmanyam, A. (1998). Investor psychology and security market under-and overreactions. *the Journal of Finance*, 53(6):1839–1885.
- David, A. and Veronesi, P. (2004). Inflation and earnings uncertainty and volatility forecasts. *Manuscript, Graduate School of Business, University of Chicago*.
- Dewachter, H. and Lyrio, M. (2006). Macro factors and the term structure of interest rates. *Journal of Money, Credit, and Banking*, 38(1):119–140.
- Diebold, F. X. and Li, C. (2006). Forecasting the term structure of government bond yields. *Journal of econometrics*, 130(2):337–364.
- Dzielinski, M. (2011). News sensitivity and the cross-section of stock returns. *NCCR Finrisk Working Paper No. 719*.
- Edison, H. J. (1996). The reaction of exchange rates and interest rates to news releases. *FEB International Finance Discussion Paper*, 2(570).
- Edmans, A., Garcia, D., and Norli, Ø. (2007). Sports sentiment and stock returns. *The Journal of Finance*, 62(4):1967–1998.
- Evans, C. L. and Marshall, D. A. (2007). Economic determinants of the nominal treasury yield curve. *Journal of Monetary Economics*, 54(7):1986–2003.

- Fama, E. F. and Bliss, R. R. (1987). The information in long-maturity forward rates. *The American Economic Review*, pages 680–692.
- Gade, T. P., Salines, M., Glöckler, G., and Strodthoff, S. (2013). 'loose lips sinking markets?': The impact of political communication on sovereign bond spreads. *ECB Occasional Paper*, 1(150).
- Garcia, D. (2013). Sentiment during recessions. *The Journal of Finance*, 68(3):1267–1300.
- Gilbert, T., Scotti, C., Strasser, G., and Vega, C. (2010). Why do certain macroeconomic news announcements have a big impact on asset prices. In *Applied Econometrics and Forecasting in Macroeconomics and Finance Workshop*, Federal Reserve Bank of St. Louis.
- Goldberg, L. S. and Grisse, C. (2013). Time variation in asset price responses to macro announcements. Technical report, National Bureau of Economic Research.
- Goldberg, L. S. and Leonard, D. (2003). What moves sovereign bond markets? the effects of economic news on us and german yields. *Current issues in economics and finance*, 9(9).
- Gross-Klussmann, A. and Hautsch, N. (2011). When machines read the news: Using automated text analytics to quantify high frequency news-implied market reactions. *Journal of Empirical Finance*.
- Gürkaynak, R. S., Sack, B., and Swanson, E. (2005). The sensitivity of long-term interest rates to economic news: evidence and implications for macroeconomic models. *The American Economic Review*, 95(1):425–436.
- Hillert, A., Jacobs, H., and Müller, S. (2014). Media makes momentum. *Review of Financial Studies*, page hhu061.
- Hirshleifer, D. (2001). Investor psychology and asset pricing. *The Journal of Finance*, 56(4):1533–1597.
- Hirshleifer, D. and Shumway, T. (2003). Good day sunshine: Stock returns and the weather. *The Journal of Finance*, 58(3):1009–1032.
- Hong, H. and Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *The Journal of Finance*, 54(6):2143–2184.

- Kamstra, M. J., Kramer, L. A., and Levi, M. D. (2003). Winter blues: A sad stock market cycle. *The American Economic Review*, 93(1):324–343.
- Leinweber, D. and Sisk, J. (2011). Event driven trading and the ‘new news’. *Journal of Portfolio Management*, 38(1).
- Maier, S. R. (2005). Accuracy matters: A cross-market assessment of newspaper error and credibility. *Journalism & Mass Communication Quarterly*, 82(3):533–551.
- Mullainathan, S. and Shleifer, A. (2005). The market for news. *The American Economic Review*, 95(4):1031–1053.
- Pindyck, R. S. (1998). *Econometric Models*. JSTOR.
- Smales, L. (2014). News sentiment in the gold futures market. *Journal of Banking and Finance*, 474:99–106.
- Smales, L. (2016). News sentiment and bank credit risk. *Journal of Empirical Finance*, 303:1361–1364.
- Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *The Journal of Finance*, 62(3):1139–1168.
- Uhl, M. (2017). Emotions matter: Sentiment and momentum in foreign exchange. *Journal of Behavioral Finance*.
- Uhl, M., Pedersen, M., and Malitius, O. (2015). What’s in the news? using news sentiment momentum for tactical asset allocation. *Journal of Portfolio Management*.
- Wright, J. H. (2011). Term premia and inflation uncertainty: Empirical evidence from an international panel dataset. *The American Economic Review*, 101(4):1514–1534.

Tables and Figures

Figure 1: Overview of Reuters News Analytics Topic Codes for News Sentiment Indicators

News Sentiment Specifications			
Indicator	Political News	Debt News	Monetary News
Filtered Topics	News related to politics, judicial, law, for US and international	News related to high yield, investment grade, US corporate bonds, government and municipal debt	News related to US Federal Reserve, inflation and FED policy makers as well as international central bank policy and tendencies
Source: Thomson Reuters NewsAnalytics			

Table 1: News Sentiment Summary Statistics

Summary statistics of news sentiment				
	Political News Sentiment	Debt News Sentiment	Monetary News Sentiment	Total News Sentiment
Mean	-0.0762	0.0973	-0.1205	-0.033137
Median	-0.0629	0.0978	-0.1327	-0.037728
Maximum	0.2096	0.3032	0.3442	0.200743
Minimum	-0.3769	-0.1262	-0.4335	-0.304073
Std. Dev.	0.1303	0.0937	0.1339	0.106645
Skewness	-0.2010	-0.2964	0.5451	-0.067758
Kurtosis	2.6815	2.8228	3.5060	2.648858
Observations	134	134	134	134

Table 2: Regressions of news sentiment on term structure maturities

OLS Model Estimates									
Regression Coefficients of OLS Model, t statistics in [] below coefficients Sample (adjusted): 2003M02 2014M02 Included Observations: 133 after adjustments									
Dependent Variables	1-month US Treasury Yield (d) 1	3-month US Treasury Yield (d) 2	1-year US Treasury Yield (d) 3	2-year US Treasury Yield (d) 4	3-year US Treasury Yield (d) 5	5-year US Treasury Yield (d) 6	7-year US Treasury Yield (d) 7	10-year US Treasury Yield (d) 8	30-year US Treasury Yield (d) 9
Independent Variables									
News Sentiment	0.837304 [3.501371]***	0.835542 [5.284959]***	0.743535 [4.862618]***	0.668534 [3.561931]***	0.651383 [3.136793]***	0.551697 [2.44018]**	0.482888 [1.959396]*	0.416328 [1.835123]*	0.379636 [1.823754]*
Constant	0.018458 [0.698383]	0.018251 [1.044529]	0.014678 [0.668521]	0.011764 [0.567134]	0.651383 [0.010856]	0.00797 [0.318949]	0.005536 [0.203236]	0.004003 [0.159644]	0.002673 [0.116193]
R- Squared	0.085576	0.175742	0.152899	0.088298	0.069863	0.043478	0.028473	0.025063	0.024761
Adjusted R- Squared	0.078596	0.16945	0.146432	0.081339	0.062763	0.036176	0.021056	0.017621	0.017317
Standard Error of Regression	0.291768	0.192894	0.186562	0.228997	0.253363	0.275849	0.300688	0.276798	0.253976
Schwarz Criterion	0.432669	-0.394964	-0.461717	-0.051828	0.150397	0.32046	0.492899	0.327327	0.155237
Durbin- Watson Statistic	2.373727	1.89052	1.753053	1.9254	1.943009	1.949723	1.948268	1.905985	1.971572

*,**,*** denotes statistical significance at the 10%, 5% and 1% level, respectively

This table shows estimation results from an ordinary least squares (OLS) model. News sentiment is regressed against yields of US Treasuries across the term structure from 1-month to 30-years. The results show that news sentiment is statistically significant in explaining changes in US Treasury yields across the term structure.

Table 3: Regressions of news sentiment and macroeconomic control variables on term structure maturities

OLS Model Estimates									
Regression Coefficients of OLS Model, t statistics in [] below coefficients Sample (adjusted): 2003M02-2014M02 Included Observations: 133 after adjustments									
Dependent Variables	1-month US Treasury Yield (d) 1	3-month US Treasury Yield (d) 2	1-year US Treasury Yield (d) 3	2-year US Treasury Yield (d) 4	3-year US Treasury Yield (d) 5	5-year US Treasury Yield (d) 6	7-year US Treasury Yield (d) 7	10-year US Treasury Yield (d) 8	30-year US Treasury Yield (d) 9
<i>Independent Variables</i>									
News Sentiment	0.79693 [3.835472]***	0.770246 [4.047924]***	0.672093 [4.472771]***	0.619078 [4.204352]***	0.605213 [3.787192]***	0.489666 [2.876495]***	0.370274 [1.744996]*	0.318028 [1.474160]	0.212049 [0.864403]
ISM Manufacturing (differenced)	0.006508 [0.517714]	0.011075 [1.213460]	0.018909 [1.695651]*	0.030183 [2.404318]**	0.033197 [2.394727]**	0.035416 [2.267496]**	0.035581 [2.220634]**	0.043468 [3.390484]***	0.040778 [3.507747]***
Consumer Price Index y-o-y (differenced) (-1)	-3.551781 [-0.818551]	-0.119511 [-0.036679]	6.092513 [1.707955]*	5.126637 [1.286069]	5.423528 [1.261043]	9.351832 [1.851022]*	13.54529 [2.140048]**	7.437881 [2.115255]**	6.779632 [2.001642]**
Non-Farm Payrolls (differenced) (-1)	-2.46E-05 [-0.206568]	1.18E-05 [0.158020]	-8.10E-05 [-1.000013]	-0.000124 [-0.888395]	-0.000158 [-0.900412]	-5.42E-05 [-0.256250]	5.42E-07 [0.002478]	6.06E-05 [0.354300]	-3.69E-06 [-0.023377]
US Treasury Yield 10y - 2y (differenced) (-1)	-0.382203 [-2.640157]***	-0.322389 [-2.617935]***	-0.103579 [-1.558451]	-0.116384 [-1.769632]*	-0.105516 [-1.268940]	-0.009389 [-0.100995]	-0.025327 [-0.266810]	-0.319624 [-1.918760]*	-0.292642 [-1.537795]
Constant	0.016447 [0.927587]	0.015548 [1.016030]	0.011819 [0.695165]	0.008111 [0.449423]	0.007117 [0.371901]	0.003024 [0.158055]	-0.000228 [-0.011883]	0.001917 [0.102967]	-0.001696 [-0.096961]
R- Squared	0.15467	0.280386	0.221285	0.16893	0.147366	0.133696	0.1365	0.198512	0.177988
Adjusted R- Squared	0.121125	0.25183	0.190383	0.135951	0.113531	0.099319	0.102235	0.166707	0.145368
Standard Error of Regression	0.286012	0.183762	0.182373	0.222627	0.247	0.266819	0.288489	0.251239	0.237388
Schwarz Criterion	0.509858	-0.374927	-0.390096	0.008784	0.216572	0.370935	0.527103	0.250605	0.137186
Durbin- Watson Statistic	2.437722	2.029624	1.975995	2.096428	2.114275	2.131883	2.08899	2.038321	2.092736
*, **, *** denotes statistical significance at the 10%, 5% and 1% level, respectively									

This table shows estimation results from an ordinary least squares (OLS) model. News sentiment and macroeconomic control variables are regressed against yields of US Treasuries across the term structure from 1-month to 30-years. The results show that news sentiment is statistically significant in explaining changes in US Treasury yields particularly at the front end of the term structure, whereas some macroeconomic variables can explain longer maturities.

Table 4: Forecasting statistics for US Treasury term structure with news sentiment

Forecasting statistics for US Treasury term structure				
<i>OLS Model</i> : US Treasury yield [1-month to 30-year] (differenced), News Sentiment, ISM Manufacturing (differenced), Consumer Price Index (yoy) (differenced) (-1), Non-farm payrolls (differenced) (-1), US Treasury 10-year - US Treasury 2-year (differenced) (-1), Constant				
Maturity	Root Mean Squared Error	Mean Absolute Error	R-squared	Adj. R-squared
1-month yield	0.279436	0.175625	0.15467	0.121125
3-month yield	0.179537	0.111735	0.280386	0.25183
1-year yield	0.17818	0.115448	0.221285	0.190383
2-year yield	0.217508	0.15323	0.16893	0.135951
3-year yield	0.241321	0.181022	0.147366	0.113531
5-year yield	0.2607	0.202403	0.133696	0.099319
7-year yield	0.281856	0.219724	0.1365	0.102235
10-year yield	0.25832	0.204685	0.14187	0.107817
30-year yield	0.23193	0.181004	0.177988	0.145368

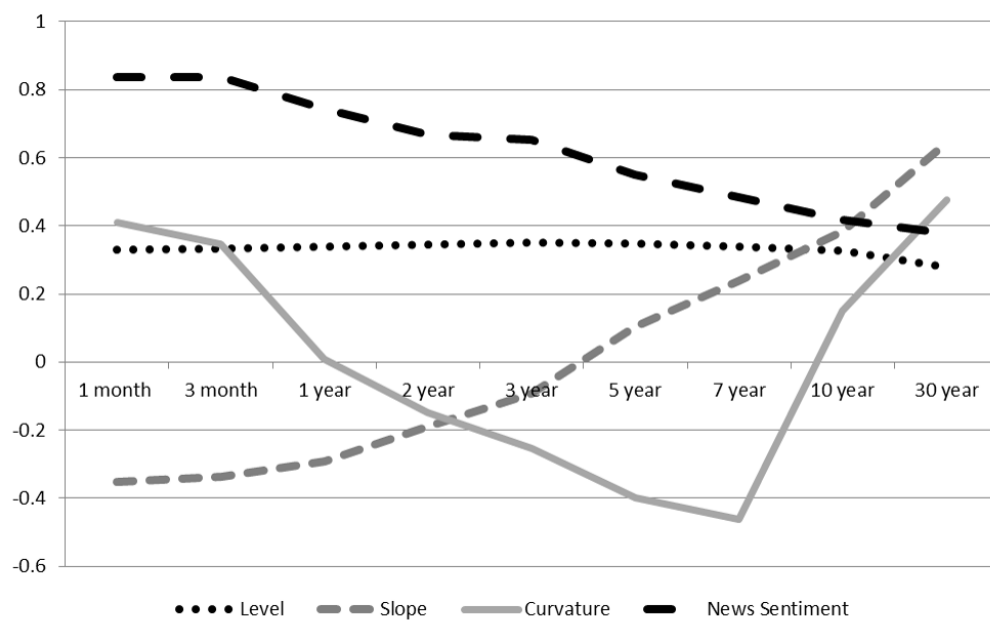
This table shows forecasting statistics, such as root mean squared error and mean absolute error as well as R-squareds, for one-step ahead forecasts from the simple OLS regressions that were performed as shown in table 2 with news sentiment, macroeconomic control variables and the yields across the term structure from 1-month to 30-year US Treasury yields.

Table 5: Forecasting statistics for US Treasury term structure without news sentiment

Forecasting statistics for US Treasury term structure				
<i>OLS Model:</i> US Treasury yield [1-month to 30-year] (differenced), ISM Manufacturing (differenced), Consumer Price Index (yoy) (differenced) (-1), Non-farm payrolls (differenced) (-1), US Treasury 10-year - US Treasury 2-year (differenced) (-1), Constant				
Maturity	Root Mean Squared Error	Mean Absolute Error	R-squared	Adj. R-squared
1-month yield	0.29122	0.179776	0.081871	0.052953
3-month yield	0.196253	0.119842	0.140146	0.113064
1-year yield	0.1915	0.12396	0.100511	0.072181
2-year yield	0.226645	0.163301	0.097644	0.069224
3-year yield	0.249227	0.187784	0.090584	0.061941
5-year yield	0.2655	0.205373	0.101332	0.073027
7-year yield	0.284419	0.223074	0.120721	0.093028
10-year yield	0.260359	0.208136	0.128266	0.10081
30-year yield	0.232954	0.180912	0.170712	0.144593

This table shows forecasting statistics, such as mean squared error and mean absolute error as well as R-squareds, for one-step ahead forecasts from the simple OLS regressions that were performed as shown in table 2 with macroeconomic variables and the yields across the term structure from 1-month to 30-year US Treasury yields.

Figure 2: Yield curve factors: level, slope, curvature and news sentiment



This figure shows the “traditional” yield curve factors, such as level, slope and curvature as outlined in the literature, such as in Cochrane and Piazzesi (2002). Furthermore, the figure shows the newly identified factor “news sentiment”, which is negatively correlated to the slope factor.

Table 6: OLS Model Estimates with News Sentiment and Fama-Bliss discount bond yields

OLS Model Estimates					
Regression Coefficients of OLS Model, HAC standard errors & covariance according to Newey-West, t statistics in [] below coefficients					
Sample (adjusted): 2003M02 2014M02					
Included Observations: 133 after adjustments					
<i>Dependent Variables</i>	1-year Fama Bliss Discount Bond Yield (diff) 1	2-year Fama Bliss Discount Bond Yield (diff) 2	3-year Fama Bliss Discount Bond Yield (diff) 3	4-year Fama Bliss Discount Bond Yield (diff) 4	5-year Fama Bliss Discount Bond Yield (diff) 5
<i>Independent Variables</i>					
News Sentiment	0.792975 [3.592685]***	0.696041 [3.131115]***	0.670434 [3.090709]***	0.648918 [2.590618]**	0.608411 [2.732461]***
Constant	0.016593 [0.932172]	0.011943 [0.646258]	0.010252 [0.519444]	0.009252 [0.459720]	0.008053 [0.393185]
R- Squared	0.164821	0.096871	0.076721	0.060839	0.053037
Adjusted R- Squared	0.158446	0.089977	0.069673	0.05367	0.045808
Standard Error of Regression	0.190283	0.226553	0.247926	0.271785	0.27405
Schwarz Criterion	-0.422223	-0.07329	0.107012	0.290776	0.307375
Durbin- Watson Statistic	1.716437	1.838291	1.855403	1.896349	1.93169
*, **, *** denotes statistical significance at the 10%, 5% and 1% level, respectively					

This table shows regressions of Fama-Bliss excess returns on news sentiment, which is statistically significant in all cases.

Table 7: Regression results from news sentiment, macroeconomic variables and yield curve factors

OLS Model Estimates						
Regression Coefficients of OLS Model, t statistics in [] below coefficients						
Sample (adjusted): 2003M03 2014M02						
Included Observations: 132 after adjustments						
<i>Dependent Variables</i>	News Sentiment	News Sentiment	News Sentiment	News Sentiment	News Sentiment	News Sentiment
	1	2	3	4	5	6
<i>Independent Variables</i>						
Non-farm payrolls (-1) (differenced)	0.0000825 [1.342802]					
Consumer Price Inflation (-1) (differenced)		3.090765 [1.403414]				
ISM Manufacturing Index (-1) (differenced)			0.005596 [0.657006]			
Level (-1) (differenced)				0.039947 [1.304012]		
Slope (-1) (differenced)					0.040643 [1.485377]	
Curvature (-1) (differenced)						0.002154 [0.290373]
Constant	-0.030598 [-1.631675]	-0.030504 [-1.680003]*	-0.030743 [-1.649850]	-0.030382 [-3.311571]***	-0.030751 [-3.359793]***	-0.030746 [-3.332126]***
R- Squared	0.011177	0.024888	0.009295	0.012911	0.016689	0.000648
Adjusted R- Squared	0.003512	0.017387	0.001674	0.005318	0.009125	-0.007039
Standard Error of Regression	0.105847	0.104716	0.10555	0.105358	0.105156	0.10601
*, **, *** denotes statistical significance at the 10%, 5% and 1% level, respectively						

This table shows the regression results from regressing macroeconomic variables on news sentiment and on the yield curve factors which were constructed as mentioned in Appendix A.2. None of the independent variables, namely non-farm payrolls, consumer price inflation, and ISM Manufacturing Index and the yield curve factors are statistically significant when regressed on news sentiment. This finding indicates that there is indeed new information portrayed in the news sentiment variable versus fundamental macroeconomic variables and the traditional yield curve factors.

Appendix

A.1 One-step ahead static forecasting

Based on the AR(1)-regression model, we perform one-step ahead out-of-sample forecasts for the entire US term structure. The forecasts are done based on the static method, meaning that for each forecast done for $t + 1$, the actual values of the variables are taken when the next forecast is performed for $t + 2$. Therefore, it follows that

$$\Delta r_t = W_t' \beta + u_t, \quad (3)$$

$$u_t = \rho_1 u_{t-1} + \epsilon_t, \quad (4)$$

where Δr_t refers to the respective US Treasury maturity yield and W_t refers to the independent variables, such as the macroeconomic variables and news sentiment, β is a vector of unknown parameters, and ϵ_t is the error term. The model is estimated with data up to $t = S - 1$. The fitted residuals are defined as $e_t = \Delta r_t - W_t' b$, where b are estimates of the unknown parameters β . Given that the values of W_t are available, the forecasts for $t = S$ are as follows:

$$\Delta \hat{r}_S = W_S' b + \rho_1 \hat{e}_{S-1}, \quad (5)$$

where the residuals $\hat{u}_t = \Delta \hat{r}_t - W_t' b$ are formed from the forecast values of Δr_t . See Pindyck (1998) for further information on the forecasting procedure.

A.2 Diebold and Li yield curve factor robustness check

In order to construct time series variables of the traditional yield curve factors “level”, “slope” and “curvature”, we use the traditional two-step approach of the Diebold-Li yield curve model according to Diebold and Li (2006). The Diebold-Li model is a variant of the Nelson-Siegel model (1987) obtained by reparamterizing the original formulation. For observation date t and time to maturity τ , the Diebold-Li model characterizes the yield $y_{t\tau}$ as the following function

$$y_t(\tau) = L_t + S_t \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) + C_t \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right), \quad (6)$$

in which L_t is the level factor, S_t is the slope factor, and C_t is the curvature factor. λ determines the maturity at which the loading on the curvature is maximized and governs the exponential decay rate of the model. Diebold and Li (2006) estimate the parameters with a two-step approach:

The first step is to fix λ and then estimate the level, slope and curvature parameters for each monthly yield curve. This process is repeated for all observed yield curves, and provides a 3-D time series of estimates of the unobserved level, slope and curvature factors. The second step is to fit a first-order autoregressive model to the time series of factors derived in the first step.

By fixing λ in the first step, what would otherwise be a non-linear least squares estimation is replaced by a relatively simple ordinary least squares (OLS) estimation. Within the Nelson-Siegel framework, it is customary to set

$\lambda = 0.0609$, which implies that the value at which the loading on the curvature is maximized occurs at 30 months.

The yield curve data that we apply consist of time series of monthly unsmoothed Fama-Bliss US Treasury zero-coupon yields as used and discussed in Diebold and Li (2006), and as described in our data section for the respective maturities.